

# ShopSphere

## S19 - Admin Dashboard and Operational KPIs

Backend + DB

Platform-Specific Sprint Planning Document

This document intentionally defines only the behavior and delivery scope of the selected platform. It does not mix implementation details from the other platforms.

# 1. Sprint Definition

Sprint: S19 - Admin Dashboard and Operational KPIs

Platform: Backend + DB

Application: ShopSphere

Objective: Implement the backend and database boundary for Admin Dashboard and Operational KPIs, including business rules, persistence and API behavior.

This sprint is complete only when the stated use cases, acceptance criteria and tests pass for this platform.

## 2. Business Use Cases

| ID    | Use Case       | Actor | Precondition                           | Main Flow                   | Expected Result                          |
|-------|----------------|-------|----------------------------------------|-----------------------------|------------------------------------------|
| UC-01 | View dashboard | Admin | Authorized report/dashboard permission | Select range, fetch metrics | Accurate operational KPIs are displayed. |

## 3. Detailed Implementation Plan

1. Implement KPI APIs for sales, orders, customers, products, pending orders, returns and refunds.
2. Support date ranges and role restrictions.
3. Optimize aggregate queries.

## 4. Data / State / Contract Scope

- dashboard aggregate queries

### API / Integration Contract

- Feature APIs for this sprint, versioned under /api/v1.
- Every endpoint must define authentication, authorization, validation, error codes and request correlation.

## 5. Rules and Expected Behavior

- The platform must rely on the backend as the source of truth for business rules.
- Do not expose stack traces, secrets or internal implementation details to end users.
- Handle loading, empty, validation and error states explicitly.
- Requests that can be retried must be safe from duplicate side effects where the feature requires idempotency.
- Ownership and authorization must be checked server-side for protected data.

## 6. Acceptance Criteria

- Data model supports the required business relationships and constraints.
- API enforces authentication, authorization, validation and ownership where applicable.
- Business rules are server-authoritative.
- Transactions/idempotency/concurrency controls are used where the feature requires them.
- Controlled error codes and request correlation are implemented.

## 7. Test Cases

| TC        | Scenario           | Expected Result                                                                                               |
|-----------|--------------------|---------------------------------------------------------------------------------------------------------------|
| TC-S19-01 | No data            | Operation is handled according to the defined business rule; no incorrect state or security leak is produced. |
| TC-S19-02 | Large date range   | Operation is handled according to the defined business rule; no incorrect state or security leak is produced. |
| TC-S19-03 | Invalid date range | Operation is handled according to the defined business rule; no incorrect state or security leak is produced. |

| TC        | Scenario          | Expected Result                                                                                               |
|-----------|-------------------|---------------------------------------------------------------------------------------------------------------|
| TC-S19-04 | Unauthorized role | Operation is handled according to the defined business rule; no incorrect state or security leak is produced. |
| TC-S19-05 | Slow aggregation  | Operation is handled according to the defined business rule; no incorrect state or security leak is produced. |
| TC-S19-06 | Timezone boundary | Operation is handled according to the defined business rule; no incorrect state or security leak is produced. |

## 8. Sprint Exit Checklist

- Implementation completed only for this platform.
- All primary and negative use cases pass.
- API contracts are implemented/consumed as defined.
- Validation and authorization tests pass.
- Loading/empty/error handling is verified where applicable.
- No critical or blocker defect remains.
- Automated tests pass in CI.
- Documentation and contract changes are committed.

## 9. Out of Scope for This Document

- Implementation behavior of Customer Portal. That behavior belongs in the separate S19 document for that platform.
- Implementation behavior of Admin Portal. That behavior belongs in the separate S19 document for that platform.